Krishnaa Vadivel | Software Engineer | Scientific Computing | HPC Workflows

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Summary

Software engineer and Yale PhD researcher building scientific-computing, automation, and high-performance workflows in Python and C++. Experience spans distributed HPC, CUDA/OpenGL tooling, containerized development, numerical software, and production support around both research and engineering systems.

Experience

Yale University, Electrical Engineering
PhD Research Assistant *2021–present*

- Develop Python/C++ workflows for first-principles and many-body simulation, using MPI, OpenMP, PETSc, SLEPc, CMake, and Linux-based automation across leadership-class HPC environments.
- Build reusable tooling around distributed compute, scientific pipelines, and ML-assisted workflows for materials and molecular problems.
- Work across research code, documentation, and theory-to-implementation translation for scalable computational workflows.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026, demonstrating research output alongside production-style software work.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Built CUDA/OpenGL software for waveform processing, interactive visualization, and diagnostic tooling for large engineering datasets.
- Developed ASP-style services and Microsoft SQL-backed workflows for internal engineering and field-data systems.
- Contributed embedded-adjacent engineering support, giving strong systems-level context for software used alongside deployed hardware.

FindLight
Photonics Technical Writer *2018–2019*

- Wrote technical content translating specialized optics and photonics topics into clear engineering-facing explanations.

Selected Projects

DOLFINx Semiconductor Workflow: Python finite-element and PETSc-based workflow for Poisson and drift-diffusion style simulation.

Agentic Scientific Automation: Retrieval-driven tooling that reads documentation and papers to generate structured computational-science workflow inputs.

Distributed Scientific Tooling: Reusable software workflows for simulation setup, engineering analysis, and container-oriented research infrastructure.

Scientific Visualization Tooling: CUDA/OpenGL and Python-based utilities for inspecting and communicating large numerical datasets.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, SQL
Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib
Systems: Linux, CMake, Docker, Docker Compose, Kubernetes, gcloud
Platforms: CUDA, OpenGL, Frontier, Frontera, Perlmutter
ML / Automation: PyTorch, scikit-learn, agentic workflows, scientific automation